

In []: Roll No:CO-2105 and Name:Sandhya Raghunath Magadum

In []: 1) int -> float , str , bool , complex

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In [1]: a=2
print("Original:",a)
print("Type:",type(a))
print("Float:",float(a))
print("String:",str(a))
print("Bool:",bool(a))
print("Complex:",complex(a))
```

Original: 2
 Type: <class 'int'>
 Float: 2.0
 String: 2
 Bool: True
 Complex: (2+0j)

In []: 2) float -> int , str , bool , complex

```
In [2]: a = 12.8
print("Original Value :", a)
print("Original Type :", type(a))
print("\nfloat to int      :", int(a))
print("Type                :", type(int(a)))
print("\nfloat to str        :", str(a))
print("Type                :", type(str(a)))
print("\nfloat to bool       :", bool(a))
print("Type                :", type(bool(a)))
print("\nfloat to complex    :", complex(a))
print("Type                :", type(complex(a)))
```

Original Value : 12.8
 Original Type : <class 'float'>

float to int : 12
 Type : <class 'int'>

float to str : 12.8
 Type : <class 'str'>

float to bool : True
 Type : <class 'bool'>

float to complex : (12.8+0j)
 Type : <class 'complex'>

In []: 3) str -> int , float , bool , complex

```
In [3]: a = "25"
print("Original Value :", a)
print("Original Type :", type(a))
print("\nstr to int      :", int(a))
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print("Type      :", type(int(a)))
print("\nstr to float  :", float(a))
print("Type      :", type(float(a)))
print("\nstr to bool    :", bool(a))
print("Type      :", type(bool(a)))
print("\nstr to complex :", complex(a))
print("Type      :", type(complex(a)))

```

Original Value : 25

Original Type : <class 'str'>

str to int : 25
Type : <class 'int'>

str to float : 25.0
Type : <class 'float'>

str to bool : True
Type : <class 'bool'>

str to complex : (25+0j)
Type : <class 'complex'>

In []: 4) bool -> int , float , str , complex

```

In [4]: a = True
print("Original Value :", a)
print("Original Type  :", type(a))
print("\nbool to int    :", int(a))
print("Type          :", type(int(a)))
print("\nbool to float   :", float(a))
print("Type          :", type(float(a)))
print("\nbool to str      :", str(a))
print("Type          :", type(str(a)))
print("\nbool to complex :", complex(a))
print("Type          :", type(complex(a)))

```

Original Value : True

Original Type : <class 'bool'>

bool to int : 1
Type : <class 'int'>

bool to float : 1.0
Type : <class 'float'>

bool to str : True
Type : <class 'str'>

bool to complex : (1+0j)
Type : <class 'complex'>

In []: 5) complex -> int , float , str , bool

```

In [5]: a = 5 + 3j
print("Original Value :", a)

```

```
print("Original Type :", type(a))
# Complex to int
# print(int(a))      # Error
# Complex to float
# print(float(a))    # Error
print("\ncomplex to str :", str(a))
print("Type           :", type(str(a)))
print("\ncomplex to bool :", bool(a))
print("Type           :", type(bool(a)))
```

Original Value : (5+3j)

Original Type : <class 'complex'>

complex to str : (5+3j)

Type : <class 'str'>

complex to bool : True

Type : <class 'bool'>

In []: